

ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 4: The Electron — Shell Mode and Charge Shell

Version: 3.0 Date: July 1, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: D-05 Harmonic Shell, B-06b Four Views, E-03 Surface, C-311 Electric-Magnetic Duality, C-309 Friction Limit, A-10 Persistent Mode, C-08 Spin-half, C-317 Boundary-Tension Weave, C-318 Four-Interaction Mass-Effect Response Status: GREEN (structure) / YELLOW (charge shell derivation incomplete)

Gray — Standard Model Reference

The electron is a fundamental lepton with no known internal structure. Mass: $0.511 \text{ MeV}/c^2$. Charge: -1 . Spin: $1/2$. The electron is a point particle in the Standard Model — no spatial extent. Electrons occupy quantized orbitals around nuclei described by quantum numbers n, l, m, s . The Pauli exclusion principle: no two electrons can occupy the same quantum state. Electron orbitals are probability distributions, not classical orbits. The electron's magnetic moment is slightly larger than predicted by Dirac theory — this anomalous magnetic moment is one of the most precisely measured quantities in physics (g -factor = $2.00231930436\dots$).

Standard Model strength: electron behavior predicted to extraordinary precision. Standard Model limitation: why the electron has the mass it does is not explained. The point-particle assumption leads to infinite self-energy (renormalization required).

2D One-Wave Interpretation

In 2D, the electron is a shell mode — a standing wave pattern on the surface of a boundary.

It is not a particle orbiting a nucleus. It is a stable pressure pattern living on the harmonic shell (D-05) of the atom.

The shell condition in 2D: $2\pi R = n\lambda$ The electron lives where the wave fits perfectly around the boundary circumference.

At $n=1$: the ground state. Smallest stable shell. At $n=2$: first excited state. Larger shell, higher energy.

The electron does not jump between orbits. It transitions between shell modes — stable standing wave patterns at different harmonic distances from the nucleus.

The charge of the electron in 2D is the pressure asymmetry at its shell boundary. The inside of the shell has higher pressure than the outside. That asymmetry is what we measure as negative charge.

3D One-Wave Interpretation

In 3D, the electron is a spherical Persistent Mode living on a harmonic shell.

The electron is not a point. It is a shell mode. It has spatial extent — the shell radius $R_n = n\lambda/(2\pi)$. The shell is the boundary where the electron's oscillation fits exactly.

The charge shell: The boundary roll-off (B-06b) at the electron shell produces a pressure cushion. That pressure cushion is the electric field of the electron. The outward-pointing pressure gradient is negative charge. The rotation of that cushion is the electron's magnetic moment.

$E_{\text{vec}} \sim -\nabla P_{\text{cushion}}$ (negative: outward pressure = negative charge) $B_{\text{vec}} \sim \nabla \times P_{\text{cushion}}$ (rotational component = magnetic moment)

Spin-half: The electron shell mode undergoes the same *4pi closure as all spin-half objects (C-08)*. $\psi(\theta + 2\pi) = -\psi(\theta)$ Two rotations to return to original state. This is not a mysterious quantum property. It is the geometry of the boundary roll-off at the shell.

The anomalous magnetic moment: The electron's g-factor is slightly larger than 2. In One-Wave: the small correction comes from the pressure cushion interacting with the background lattice field. The cushion is not perfectly symmetric — the lattice has a small residual coupling that adds a tiny additional rotational component. This is the One-Wave analog of QED loop corrections. Formal derivation deferred.

The Pauli exclusion principle: Two electron shell modes cannot occupy the same harmonic shell with the same phase orientation. In One-Wave: two modes with identical phase orientation at the same shell would interfere destructively (F-05 Interference, $\phi = \pi$). The shell collapses. Exclusion is a consequence of destructive interference, not an independent postulate.

Mathematics

Harmonic shell condition (D-05): $2\pi R_n = n\lambda \quad R_n = n\lambda / (2\pi)$

Ground state: $R_1 = \lambda / (2\pi)$ First excited: $R_2 = 2\lambda / (2\pi) = \lambda / \pi$

Shell energy (form — Yellow): $E_n \sim \hbar \omega_n$ (Proportionality — constant of proportionality deferred to CCD-01 resolution)

Charge from pressure cushion (B-06b): $\Delta E = E_{\text{mode}} - \sigma A_s \Delta E > 0 \Rightarrow \text{roll-off} \Rightarrow P_{\text{cushion}} = \beta \Delta E / V_c$ $E_{\text{vec}} \sim -\nabla P_{\text{cushion}}$ (outward = negative) $B_{\text{vec}} \sim \nabla \times P_{\text{cushion}}$ (rotational = magnetic moment)

Spin-half closure (C-08): $\psi(\theta + 2\pi) = -\psi(\theta) \quad \psi(\theta + 4\pi) = \psi(\theta)$

Pauli exclusion from destructive interference (F-05): Two modes at same shell, same phase: $\phi = \pi \Rightarrow \psi_T = 0 \Rightarrow \text{destructive cancellation} \Rightarrow \text{exclusion}$

Anomalous magnetic moment (sketch — Yellow): $g = 2 + \delta g$ $\delta g \sim \alpha / \pi$ (α = fine structure constant) One-Wave analog: δg from residual lattice coupling at shell boundary. Derivation deferred.

Predictions

1. The electron has spatial extent equal to the ground state harmonic shell radius. Prediction: $R_{\text{electron}} = \lambda / (2\pi)$ at $n=1$. This gives a measurable lower bound on electron size, contradicting the point-particle assumption.
2. Electron charge is a pressure geometry property, not an intrinsic quantum number. Prediction: charge magnitude is proportional to the pressure cushion amplitude at the shell boundary. Electrons and positrons have equal and opposite pressure cushions.
3. Pauli exclusion follows from destructive interference (F-05). Prediction: two identical shell modes at the same harmonic shell will always show destructive interference — no additional postulate needed.
4. Electron spin follows from boundary roll-off *4pi closure* (B-06b, C-08). Prediction: the g-factor correction (anomalous magnetic moment) comes from residual lattice coupling at the shell boundary.
5. The electron Mass Effect must come from the complete four-interaction response of the stable electron candidate. The ground-state shell geometry contributes to the electrical-shell and

Boundary-Tension terms, but it cannot supply the mechanism alone. Prediction: one fixed work metric must produce

$$m_{e,\text{eff}} = \frac{1}{3} \text{Tr } \mathcal{M}_e$$

and match the measured $0.511 \text{ MeV}/c^2$ comparison value without inserting that value into the coefficients.

Yellow Audit

- Harmonic shell radius R_1 derivation requires λ measurement (CCD-01)
 - Charge shell derivation from pressure cushion not yet fully completed (B-06b)
 - Anomalous magnetic moment derivation from lattice coupling deferred
 - Pauli exclusion from destructive interference needs formal proof (F-05)
 - Electron four-interaction Mass-Effect tensor not yet derived or calibrated
 - Shell mode spatial extent prediction testable but not yet tested
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Future Work

Measure λ via Wave Reader. Derive R_{electron} from $n=1$ harmonic shell condition. Complete charge shell derivation from B-06b pressure cushion. Derive Pauli exclusion formally from F-05 destructive interference. Model anomalous magnetic moment from lattice coupling correction. Compare predicted electron radius to experimental upper bounds.

Closing Thoughts

The electron is not a point. It is a shell mode — a standing wave that fits perfectly around the boundary of its harmonic shell.

The infinite self-energy problem that plagues the Standard Model point-particle electron dissolves when the electron has spatial extent. You cannot squeeze infinite energy into a shell mode because the shell has a minimum size set by the lattice wavelength.

Renormalization was a fix for a problem that only exists if you assume particles are points. One-Wave does not assume points. The electron has a floor. The infinities go away.

END OF BOOK 1 CHAPTER 4 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.